Supplementary Material for "OpenBilliards: a validated, bit-exact event-driven simulator for cue-sports mechanics, with a systematic assessment of the empirical laws of pool"

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Abstract

This document is the online supplement to the main article. Section S1 maps every figure and table of the paper to its experiment, pinned output hash, and registered source. Section S2 states the tier-1 reproduction mechanics of the artifact bundle. Section S3 reproduces the governing equations and frozen constants of every model-catalog row. Section S4 carries the extended per-experiment result prose for experiment groups B (stick–ball) and D (curved paths), whose summaries appear in the main text. Section S5 transcribes the full predicted-shift verification table for the slip-law and branch-selection fix cascade. Section S6 documents the presentation methods behind the artifact bundle's validated-scenario video pack — the C^2 log-domain time remap and shutter synthesis — which render already-verified kernel records and carry no validation claims. Numbered cross-references to sections, tables, and figures without an "S" prefix resolve to the main article.

S1. Data provenance: figures and tables to pinned outputs

Table S1 states the paper's reading contract: every plotted number regenerates from one registered script against the pinned kernel, and the expected sha256 of each output is recorded both in the study's baseline manifest (the post-cascade baseline of record under the canonical kernel `0.1.0+fb3425ad`) and in the artifact bundle's machine-readable expected-hash table, which is parsed from that manifest rather than hand-written. The scripts, the manifest, and the bundle are not part of the public source release (main-text Section 7.1). Slot F1 (the architecture schematic) carries no experiment data. Tables 2, 10, and 12 draw on battery output metadata, the group subsections' ledgered numbers, and the determinism evidence index, respectively.

Table S1: Provenance map: figure slot → experiment → expected output hash (prefix) → registered source.

Slot	Exp	sha256 (8)	Registered source
F3	A1	`ec64b364`	TP 3.1, A.5, A.7
F3	A2	`865815ab`	TP 3.3, A.4
F2	A3	`f1367bc1`	TP A-14, A.28, B.3
F4	A4	`dcbe8050`	TP B.25, FAQ speed-effects
F4	A5	`df91770b`	TP A.26
F4	A6	`c24e4e37`	TP A.27
F5	A7	`87bd5f07`	TP A.17, A.14 bottom
F6	B1	`faebd431`	TP A-31; Cross (2008)
F6	B2	`3b6356e4`	TP B.1, B.26
F6	B3	`1ed54d97`	TP 2.1, A.12
F6	B4	`a42720a6`	TP 4.2
F7	C1	`0b3d1246`	TP B.28, B.27

Slot	Exp	sha256 (8)	Registered source
F8	C2	`64410241`	TP 7.2; Cohen (2002)
F8	C3	`3d12bdf4`	FAQ; TP 6.3, 7.3
F9	C4	`32feed7a`	Mathavan et al. (2010) Figs 7–10; HSV B.15
F10	D1	`8a4a53a2`	TP A.4; HSV A.76; Coriolis (1835)
F10	D2	`2178b9d8`	TP A.19; Coriolis (1835)
F10	D3	`c5a5113f`	TP B.23
F11–F12	E1	`21f9efd2`	Jewett via BD Dec 2024; HSV A.88–A.95
F11	E2	`ea6cff60`	TP B.21, A.15
F12	E3	`87121a5b`	HSV 7.6/7.7, B.43, B.45
F13	F1	`28ee6348`	TP 4.1, B.5, B.6, B.8
F13	F2	`78d1b730`	TP B.8/B.9; FAQ trisect
F13	F3	`f0ee2200`	FAQ jump; HSV A.1/A.2, B.37
F14	ORACLE-X	`00a1488e`	Kiefl (2024), pinned oracle env

The abstract's battery-scale figure — 16,933 recorded simulations — is an exact, artifact-backed count, not a program-management estimate: it is the number of kernel simulations the battery executes end-to-end, recorded one row per call (serde-exact inputs plus stamp-normalized trajectory digest) by the input tap behind the wasm-replay corpus of Section 7, whose SHA-256 and per-experiment row counts are pinned in the bundle's replay report. It supersedes the pre-replay floor of "more than 4,800", which was derived from the pinned outputs alone: the baseline manifest's digest normalization counts 3,157 digest-pinned per-record trajectory fingerprints across the 25 outputs — all 3,157 of which appear in the replay corpus's digest set — and the remainder are digest-free recorded sweep rows (e.g. the 492-run A4 grid) carried in the same pinned files.

One artifact is deliberately outside this contract. The computational performance evidence of Section 3.6 (Table 3) is engineering-tier, not physics V&V: its timings are machine-dependent by nature, so its artifact (a pinned benchmark report in the bundle) records host provenance, both measurement passes, and the protocol, but carries *no* expected hash and is excluded from the physics-identity manifest. Its determinism guards are digest ties, not timing hashes: the timed kernel records reproduce the stamp-independent trajectory digests of the ORACLE-X output (30/30 matched), proving the timed scenarios are the event-verified ones.

Three run-order dependencies exist; every other script is order-free. Experiment A7 reads the A3 output; ORACLE-X reads the A3 and A6 outputs and additionally requires the pinned pooltool environment; D1 and F2 re-read only their own outputs. Digitized-source fixtures (the TP curves, the Mathavan figures, the FAQ charts) carry per-fixture provenance blocks and digitization uncertainty; no third-party source frames are redistributed.

S2. Tier-1 reproduction procedure

Artifact bundle contents.. The artifact bundle, which is not part of the public source release and has not been published (main-text Section 7.1), contains: the canonical WebAssembly binary (`pool_wasm_bg.wasm`, stamp `0.1.0+fb3425ad`, sha256 `5e3b6ee499b32c868689c3f5279bbd028161c94bdd306fbe969a2dbe5ef34936`); a binary Python wheel (abi3, CPython $\geq$ 3.9) and an npm SDK tarball; the 24 experiment scripts, the oracle harness, the shared experiment library, and a 13-check environment smoke test; the sensitivity pack (the sensitivity scripts, the report generator,

and the pinned sensitivity report) together with the physics-identity checker it imports and the checker's normalized-baseline manifest and machine waiver ledger; exactly the fixtures the scripts read, each with its provenance block; the machine-readable expected-hash table, parsed from the baseline manifest and never hand-written; the waiver ledger verbatim; the validated-scenario video pack — sixteen annotated slow-motion entries (final renders 2026-08-05), each staged strike replayed through the then-current shipped canonical binary under a gate-before-capture law (the claimed pinned record digest must reproduce before any frame is captured; the per-entry kernel stamp is recorded in the gate report, and the pack's stamps are later canonical stamps than the battery pin above, each entry's digests pinned against its own stamp), with a per-entry machine-readable gate report (the derived frame-schedule summary — frame counts, worst rate-step ratio, shutter parameters, replay spans — gate verdicts, and, for part of the pack, capture-verification records; the playback plans themselves are authored in the render tool's scenario catalog), a poster frame, and a pack ledger carrying per-entry SHA-256 hashes of the delivered bytes together with the render and verification history, including the cue-model provenance (presentation methods in Section S6; the videos present verified records and carry no claims of the paper) — and a SHA-256 manifest covering every file in the bundle.

Procedure.. Tier-1 reproduction is five steps: (1) verify the SHA-256 manifest; (2) install the wheel; (3) run the 13-check environment smoke test; (4) run any experiment script; (5) compare the output sha256 against the expected-hash table. Verification *is* hash equality: a tier-1 reproduction suc-

ceeds exactly when the regenerated output bytes hash to the pinned value.

Checker semantics.. The physics-identity checker (normalization spec v2) masks only kernel-version stamps and compares per-record trajectory digests as stamp-independent physics evidence; 25/25 outputs pass on the baseline manifest of record.

Reproducibility caveats, stated honestly.. Wheel bytes are not guaranteed bit-reproducible across machines and toolchains — the determinism obligation attaches to *outputs*, not to package bytes. Raw output hashes embed the kernel stamp, so a differently-stamped wheel requires the documented normalization until the submission-freeze rebuild.

Licensing.. The kernel source is public under the MIT license, and the portions derived from pooltool remain under the Apache License 2.0 (main-text Section 7.1); the bundle's license terms will be stated when it is published. The bundle redistributes no Dr. Dave frames or other third-party captures — only derived measurement tables with provenance.

S3. Governing equations and frozen constants of the validated models

This appendix makes each row of Table 2 reproducible from the paper alone. Every equation is stated *as implemented at kernel* `0.1.0+fb3425ad` (the canonical binary of Section 3.4); the citation attached to each model identifies its published lineage, and wherever the implementation deviates from the cited source the deviation is itemized in place rather than left behind the word "corrected." Formulations inherited from the open-source reference

implementation pooltool (Kiefl, 2024) are cited as such; the oracle comparison of Section 6 independently verifies that inheritance event-by-event *on the matched registry only*—the 38 registered A3/A6 points, which exercise the Han cushion family, the legacy ball–ball law, and matched strike models. The Stronge, Mathavan, and compliant-cluster rows are never oracle-verified (Section 6 states this explicitly for Stronge); their evidence is the published-curve anchors and battery gates of Sections 5 and 6. All frozen constants appear in Table S2 with units and provenance.

Notation: $\mathbf{v}$ and $\boldsymbol{\omega}$ are a ball's center-of-mass velocity and angular velocity, R its radius, m its mass, $I = \frac{2}{5}mR^2$ its moment of inertia, $\hat{\mathbf{z}}$ the vertical unit vector, and $g = 9.81\,\mathrm{m/s^2}$. Subscripts n and t denote components along a contact normal $\hat{\mathbf{n}}$ and tangent $\hat{\mathbf{t}}$. A hat denotes a unit vector; a prime denotes a post-impact quantity.

S3.1. Ball–cloth motion between events

As implemented at kernel `0.1.0+fb3425ad`, cloth contact uses three independent friction regimes—sliding μ_s, rolling μ_r, and vertical-spin decay μ_{sp}—with closed-form kinematics in each regime, following the classical treatment collected in the technical-proof corpus (Alciatore, n.d.c) and the event-based formulation of Leckie and Greenspan (2006) as realized by Kiefl (2024). The cloth contact-point slip velocity is

$$\mathbf{u} \;=\; \mathbf{v} + R\,(\hat{\mathbf{z}} \times \boldsymbol{\omega}). \tag{S1}$$

Sliding. While $\mathbf{u} \neq \mathbf{0}$, planar Coulomb friction acts along the (exactly constant) initial slip direction $\hat{\mathbf{u}}_0$:

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t - \tfrac{1}{2}\,\mu_s g\, t^2\, \hat{\mathbf{u}}_0, \qquad \mathbf{v}(t) = \mathbf{v}_0 - \mu_s g\, t\, \hat{\mathbf{u}}_0, \tag{S2}$$

$$\boldsymbol{\omega}_{xy}(t) = \boldsymbol{\omega}_{0,xy} - \frac{5\,\mu_s g}{2R}\, t\, (\hat{\mathbf{u}}_0 \times \hat{\mathbf{z}}), \qquad \mathbf{u}(t) = \mathbf{u}_0 - \tfrac{7}{2}\,\mu_s g\, t\, \hat{\mathbf{u}}_0. \tag{S3}$$

The constancy of $\hat{\mathbf{u}}_0$ is an exact consequence of planar Coulomb friction, not an approximation; the slide ends when $\mathbf{u} = \mathbf{0}$, at

$$\tau_{\text{slide}} = \frac{2\,|\mathbf{u}_0|}{7\,\mu_s g}. \tag{S4}$$

Rolling. A rolling ball travels in a straight line along $\hat{\mathbf{v}}_0$ with constant deceleration $\mu_r g$, and the rolling constraint is re-imposed exactly at every instant (not integrated):

$$\mathbf{v}(t) = \mathbf{v}_0 - \mu_r g\, t\, \hat{\mathbf{v}}_0, \qquad \boldsymbol{\omega}_{xy}(t) \;=\; \frac{\hat{\mathbf{z}} \times \mathbf{v}(t)}{R}, \qquad \tau_{\text{roll}} = \frac{|\mathbf{v}_0|}{\mu_r g}. \tag{S5}$$

Vertical spin. ω_z decays linearly and independently in every *cloth-contact* moving state (sliding, rolling, spinning), clamped at exactly zero (no overshoot)—airborne flight is excluded: it freezes all of $\boldsymbol{\omega}$ (below), so a ball in flight lands with the spin it launched with:

$$\omega_z(t) = \operatorname{sgn}(\omega_{z,0})\, \max\big(|\omega_{z,0}| - \alpha t,\; 0\big), \qquad \alpha = \frac{5\,\mu_{sp}\, g}{2R}, \tag{S6}$$

with the frozen $\mu_{sp} = \frac{4}{9}R$ giving $\alpha = \frac{10}{9}g \approx 10.9\,\text{rad/s}^2$. A units convention governs this line: μ_{sp} is *dimensionless* — the inherited shorthand $\frac{4}{9}R$ consumes the numeric value of R in metres, i.e. $\mu_{sp} = \frac{4}{9}\,(R/1\,\text{m}) = 0.0127$ — so $\alpha = 5\mu_{sp}g/(2R)$ carries s^{-2} as required; both the coefficient and the resulting rate are printed so the implementation is reproducible either way. The spin regime ends at $\tau_{\text{spin}} = |\omega_z|/\alpha$; a ball is projected to exact rest (all components identically zero) when its last regime expires.

Airborne flight. A ball off the bed follows an exact ballistic parabola, $\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t - \frac{1}{2} g t^2 \hat{\mathbf{z}}$, with $\boldsymbol{\omega}$ frozen: the design excludes air drag and Magnus lift so that every trajectory stays polynomial (Section 3.1). Every landing resolves through the slate impact of S3.2.

S3.2. Ball–slate impact (landing resolution)

As implemented at kernel `0.1.0+fb3425ad`, every landing of an airborne ball on the bed—including the immediate post-strike bounce that every jump shot dispatches to at $\tau \approx 0$ (S3.4)—resolves through a frictional inelastic impulse in the formulation of Kiefl (2024). The ball is first re-seated to exactly $z = R$. With $v_z < 0$ the incoming vertical velocity, the normal channel is restitution e_t:

$$v_z' = -e_t\, v_z \qquad \text{(normal impulse } (1+e_t)\,|v_z| \text{ per unit mass).} \tag{S7}$$

The tangential channel is resolved with the normal problem settled first: the cloth contact slip $\mathbf{u} = \mathbf{v}_{xy} + R\,(\hat{\mathbf{z}} \times \boldsymbol{\omega})$ (Eq. (S1), planar velocity only) selects between the Coulomb slip impulse and the 2/7-stick impulse, stick applying when there is no slip or the stick impulse is not larger in magnitude:

$$\Delta\mathbf{V}_t = \begin{cases} -\,\mu_s\,(1+e_t)\,|v_z|\,\hat{\mathbf{u}}, & \mathbf{u} \neq \mathbf{0} \text{ and } |\Delta\mathbf{V}_{t,\text{stick}}| > |\Delta\mathbf{V}_{t,\text{slip}}| \quad \text{(slip)}, \\ \frac{2}{7}\big[R\,(\boldsymbol{\omega} \times \hat{\mathbf{z}}) - \mathbf{v}_{xy}\big], & \text{otherwise (stick)}, \end{cases} \tag{S8}$$

with the corresponding spin updates

$$\text{slip:}\quad \boldsymbol{\omega}' = \boldsymbol{\omega} + \frac{5}{2R}\,|\Delta\mathbf{V}_t|\,(\hat{\mathbf{z}} \times \hat{\mathbf{u}}), \qquad \text{stick:}\quad \boldsymbol{\omega}' = \tfrac{2}{7}\,\boldsymbol{\omega} + \tfrac{5}{7}\,\frac{\hat{\mathbf{z}} \times \mathbf{v}_{xy}}{R} \tag{S9}$$

(note the stick update also scales ω_z by 2/7). That ω_z scaling deserves an explicit acknowledgment rather than a parenthesis: a tangential impulse

applied at the contact point directly below the ball centre exerts no torque about the vertical axis, so the 5/7 vertical-spin loss per stuck bounce is not derivable from the printed impulse — it is an inherited artifact of the ported formulation (Kiefl, 2024), disclosed here as a known limitation: every jump shot dispatches an immediate slate bounce, so a landing that resolves through the stick branch bleeds side spin accordingly. The friction coefficient is the cloth sliding μ_s—the bounce happens through the cloth. Finally the bounce guard: if the outgoing apex $v_z'^2/2g$ falls below the keep threshold—5 mm under the parity default, 10 μm under the three realism presets, the same two-regime convention as the ball–ball micro-hop guard of S3.3—then $v_z' := 0$ and the bounce cascade terminates (the apex decays geometrically with e_t^2). The frozen $e_t = 0.5$ is the reference default (Table S2); no published measurement exists.

S3.3. Ball–ball impact

As implemented at kernel `0.1.0+fb3425ad` (catalog row `CoulombCap`), the ball–ball impact is an instantaneous impulse with a normal restitution channel and a tangential friction channel, following the throw model of the technical-proof corpus (Alciatore, n.d.d,n) in the formulation of Kiefl (2024). The contact normal $\hat{\mathbf{n}}$ is the full three-dimensional line of centers from ball 1 to ball 2.

Normal channel. With e_b the arithmetic mean of the two balls' restitution parameters ($e_b = 0.95$ for the catalog ball set), the normal velocity compo-

nents exchange as an equal-mass one-dimensional inelastic collision:

$$v'_{1n} = \tfrac{1}{2}\big[(1-e_b)\,v_{1n} + (1+e_b)\,v_{2n}\big], \qquad v'_{2n} = \tfrac{1}{2}\big[(1+e_b)\,v_{1n} + (1-e_b)\,v_{2n}\big]. \tag{S10}$$

Friction coefficient. The tangential channel samples the velocity-dependent friction curve of Alciatore (n.d.d) at the contact-point relative surface speed $u_{12} = |\mathbf{u}_{12}|$, where $\mathbf{u}_{12}$ is the relative velocity of the two contacting surface points (both balls' spins included):

$$\mu_{bb}(u_{12}) \;=\; 0.009951 + 0.108\, e^{-1.088\,u_{12}}. \tag{S11}$$

(The revision drift within the source corpus for the decay constant is ledgered as a source discrepancy, Section 3.2.)

Slip branch (Coulomb cap). If the contact is slipping, each ball receives an equal-and-opposite tangential velocity change of magnitude $\mu_{bb}\,\hat{J}_n$ opposing the slip, with the per-unit-mass normal impulse

$$\hat{J}_n \;=\; \frac{1+e_b}{2}\,\big|v_{1n} - v_{2n}\big| \qquad \text{(TP A.5, Eq. 10; catalog law } \texttt{CoulombCap}\text{)}, \tag{S12}$$

$$\Delta\mathbf{V}_{t,1} = -\,\mu_{bb}(u_{12})\,\hat{J}_n\,\hat{\mathbf{u}}_{12}, \qquad \Delta\mathbf{V}_{t,2} = -\Delta\mathbf{V}_{t,1}, \tag{S13}$$
$$\Delta\boldsymbol{\omega}_1 = \Delta\boldsymbol{\omega}_2 = \frac{5}{2R}\,\hat{\mathbf{n}} \times \Delta\mathbf{V}_{t,1}.$$

In a stun cut this yields a slip-regime throw angle of exactly $\arctan\mu_{bb}$, independent of e_b. The *legacy* law (recorded as `PostImpactNormalDelta`; retained solely for byte-exact replay of pre-correction records and oracle parity, Section 6) scales the same impulse with $e_b\,|v_{1n} - v_{2n}|$ instead, leaving slip-regime throw $\arctan[2\mu_{bb}e_b/(1+e_b)]$—about 2.6% low at $e_b = 0.95$ (the slip-law defect).

Stick branch and boundary. If the contact has no initial slip, or the slip solution of Eq. (S13) would reverse the slip direction ($\mathbf{u}_{12} \cdot \mathbf{u}'_{12} \leq 0$), the no-slip ("1/7") solution applies instead:

$$\Delta\mathbf{V}_{t,1} = -\tfrac{1}{7}\Big[(\mathbf{v}_1 - \mathbf{v}_2)_t + R\,(\boldsymbol{\omega}_1 + \boldsymbol{\omega}_2) \times \hat{\mathbf{n}}\Big], \tag{S14}$$
$$\Delta\boldsymbol{\omega}_{1,2} = -\tfrac{5}{14}\Big[\frac{\hat{\mathbf{n}} \times (\mathbf{v}_1 - \mathbf{v}_2)}{R} + (\boldsymbol{\omega}_1 + \boldsymbol{\omega}_2)_t\Big].$$

The stick/slip boundary is therefore implicit, $u_{12} = 7\,\mu_{bb}(u_{12})\,\hat{J}_n$, and moves with the cap: the corrected law's boundary cut sits 0.6–1° above the legacy law's (the predicted-shift verification of Section 6). Spin components along $\hat{\mathbf{n}}$ are unchanged by the impact.

Vertical micro-hop guard. The three-dimensional line of centers gives the friction impulse a vertical component (ball hop). An on-bed ball keeps its post-impact v_z only when the resulting bounce apex $v_z^2/2g$ clears a threshold: 5 mm under the parity default (absorbing the micro-hop, matching the planar reference formulation) and 10 μm under the three realism presets of Table 2 (`stronge_realistic`, `mathavan_fitted`, `grip_fitted`), which keep the physical micro-hop while absorbing numerical dust.

S3.4. Stick–ball strike

As implemented at kernel `0.1.0+fb3425ad`, the strike is the instantaneous-point cue impact of the technical-proof corpus (TP A-30) with the endmass squirt model (TP A-31) (Alciatore, n.d.e; Cross, 2008; Shepard, 2001), in the formulation of Kiefl (2024). Shot parameters are the cue speed $v_0 \in (0, 15]$ m/s, azimuth φ, elevation $\theta \in [0, \pi/2)$, and the commanded *tip-center* offsets $(a_{\mathrm{cmd}}, b_{\mathrm{cmd}})$ in units of R (side, vertical),

subject to the miscue envelope

$$a_{\mathrm{cmd}}^2 + b_{\mathrm{cmd}}^2 \;\le\; 0.25 \qquad (\text{offset radius} \le R/2); \tag{S15}$$

shots beyond the envelope are rejected as invalid input, never resolved. Post-registration, a kernel amendment (kernel `0.1.0+8efd660e`) re-binds this limit to the *contact* offset — the variable in which the 0.5–0.55R literature bracket is stated — raising the commanded cap to $0.5\,(1 + \rho/R)$ ($0.686R$, nickel tip); every previously accepted input reproduces bit-exactly (B3's registered nickel-tip rejection probes, which would now be accepted, are the one disclosed behavioral change class), so Eq. (S15) is the envelope of record for every result in this paper (amendment note: main-text Section 8).

Commanded-to-internal offset map (frame and sign). Two side-offset conventions exist and must not be conflated. The *commanded* offset a_{cmd} of the shot record is positive for *right* English (the shooter's perspective); the *internal* offset a consumed by every formula below is measured toward the shooter's *left* ($+\hat{\mathbf{x}}$ in the cue-aligned frame). The resolver's first act is therefore a sign inversion, fused with the tip-curvature contact correction (tip-center offsets map to the contact point on the ball surface through the tip curvature radius ρ):

$$a \;=\; \frac{-\,a_{\mathrm{cmd}}}{1 + \rho/R}, \qquad b \;=\; \frac{b_{\mathrm{cmd}}}{1 + \rho/R}, \tag{S16}$$

with $\rho = 0.0106\,\mathrm{m}$ (nickel) for the default cue, and $c = \sqrt{1 - a^2 - b^2}$ is the contact-normal component (offsets internal and tip-corrected hereafter). A reproduction that scales but does not negate the side offset mirrors both the spin of Eq. (S18) and the squirt direction of Eq. (S19) on every side-English shot.

Ball speed. With cue mass M, ball mass m, and tip–ball restitution e_{tip}, the post-strike ball speed is

$$v \;=\; \frac{(1+e_{\text{tip}})\,v_0}{1+\dfrac{m}{M}+\dfrac{5}{2}\,(a^2+b^2)}, \tag{S17}$$

the elevation-independent (ball-frame) form of TP A-30; the frozen default $e_{\text{tip}} = 1$ reproduces the classical $2v_0$ numerator, and the cue catalog carries measured tip values (leather 0.73, phenolic 0.85; Table S2). In the cue-aligned frame (cue axis along $-\hat{\mathbf{y}}$, elevated by θ about $\hat{\mathbf{x}}$; a measured toward $+\hat{\mathbf{x}}$, the shooter's left; b upward) the outgoing velocity is $(0, -v\cos\theta, -v\sin\theta)$ and the angular velocity is

$$\boldsymbol{\omega}_{\text{cue}} \;=\; \frac{5v}{2R}\,\big(b,\; a\sin\theta,\; -a\cos\theta\big), \tag{S18}$$

both rotated to the table frame by $\varphi + \pi/2$ about $\hat{\mathbf{z}}$. Elevation drives the ball downward ($v_z = -v\sin\theta$); for $0 < \theta < \theta_{\text{full}}$ the ball leaves the strike airborne—the pinch of Eq. (S20) removes only the fraction $\lambda < 1$ of the downward component—and its immediate slate bounce (S3.2) is the legal-jump mechanism. At full pinch, $\theta \geq \theta_{\text{full}}$, $\lambda = 1$ and the bed reaction removes the *entire* downward component before motion-state classification: the strike exits grounded, with no slate bounce.

Squirt (cue-ball deflection). The endmass model of TP A-31 (eq. 13 there; cf. Shepard, 2001; Cross, 2008) rotates the outgoing *velocity* (spin untouched) away from the aiming line by the squirt angle ψ, directed opposite the applied side spin:

$$\tan\psi \;=\; \frac{\frac{5}{2}\,a\,\sqrt{1-a^2}}{1+\dfrac{m}{m_e}+\dfrac{5}{2}\,(1-a^2)}, \tag{S19}$$

with m_e the effective endmass. The default cue freezes $m_e = m/30 \approx 5.67\,\mathrm{g}$; the per-shaft catalog spans 1.70–12.7 g through the small-offset pivot-length inversion $m/m_e = 2.5\,L_p/R - 3.5$ of Shepard (2001), with L_p the robot-measured pivot length (Table S2).

Pinch (elevated-strike) regime. Below $\theta_{\mathrm{onset}} = 60°$ the strike is exactly Eqs. (S17)–(S19). Above it the tip pins the ball against the cloth (the grounded-massé regime of Coriolis, 1835) and the bed reaction participates in the strike, with engagement fraction

$$\lambda(\theta) = s^2(3-2s), \qquad s = \mathrm{clamp}\Big(\frac{\theta - \theta_{\mathrm{onset}}}{\theta_{\mathrm{full}} - \theta_{\mathrm{onset}}},\, 0,\, 1\Big), \qquad \theta_{\mathrm{full}} = 70°, \tag{S20}$$

a C^1 smoothstep. The bed absorbs the normal launch impulse $\hat{\jmath} = \lambda(\theta)\,|v_z|$ per unit mass ($v_z \to (1-\lambda)v_z$), and the same impulse drags cloth friction $\mu_s\,\hat{\jmath}$ through the contact patch opposite the contact-point slip, with exactly the slip/stick selection of Eq. (S8) (normal impulse magnitude $\hat{\jmath}$ in place of $(1+e_t)|v_z|$). The pinch leaves the ball pressed into the cloth: for a window $\tau_{\mathrm{pinch}} = 0.12\,\mathrm{s}$ the sliding friction deceleration is elevated to $\mu_s k g$ with

$$k \;=\; 1 + \frac{\kappa\,\hat{\jmath}}{g\,\tau_{\mathrm{pinch}}}, \qquad \kappa = 0.45, \tag{S21}$$

each phase remaining an exact parabola. The constants $(\tau_{\mathrm{pinch}}, \kappa)$ are first-party fits to reference massé footage (Table S2).

S3.5. Ball–cushion impact

All four cushion rows of Table 2 share the contact geometry and the two velocity-response hooks; they differ in the impulse model. Throughout, h is the cushion-nose height above the bed and the nose contact angle θ_a satisfies

$$\sin\theta_a \;=\; \frac{h}{R} - 1. \tag{S22}$$

The US-pool (WPA) geometry freezes $h = 0.635 \cdot 2R = 1.27R$, so $\sin\theta_a = 0.27$; the snooker-lineage geometry of Mathavan et al. (2010) places the contact at $h = 7R/5$ ($\sin\theta_a = 0.4$) and is paired exclusively with the snooker table configuration (the domain separation of Section 3.2).

Velocity-dependent restitution hooks. Two separate restitution definitions exist and are never conflated. The *kinematic* restitution consumed by the Han and Stronge models is

$$e(v_n) \;=\; \mathrm{clamp}\big(e_0 + k_v\,(v_n - v_{\mathrm{ref}}),\; e_{\min},\; e_{\max}\big), \tag{S23}$$

with v_n the normal impact speed; the frozen default is constant ($e_0 = 0.85$, $k_v = 0$, $v_{\mathrm{ref}} = 1\,\mathrm{m/s}$, clamp $[0.6, 0.98]$). The *energetic* (Stronge-sense) restitution consumed by the Mathavan model shares the same functional form and clamps but pivots on its own base ee_0:

$$ee(v_n) \;=\; \mathrm{clamp}\big(ee_0 + k_v\,(v_n - v_{\mathrm{ref}}),\; e_{\min},\; e_{\max}\big), \tag{S24}$$

with $ee_0 = 0.98$ at the structural default (the snooker fit of Mathavan et al., 2010), $ee_0 = 0.80$ constant for `mathavan_fitted`, and the velocity-sloped `grip_fitted` response

$$ee(v_n) = \mathrm{clamp}\big(0.65 + 0.18\,(v_n - 1),\; 0.5,\; 0.98\big). \tag{S25}$$

Slip-velocity friction hooks. Cushion-nose friction is an affine function of the contact-point slip speed v_{slip} at impact start, clamped:

$$\mu_w(v_{\mathrm{slip}}) \;=\; \mathrm{clamp}\big(\mu_0 + k\,(v_{\mathrm{slip}} - v_{\mathrm{ref}}),\; \mu_{\min},\; \mu_{\max}\big). \tag{S26}$$

The default row `han2005_default` freezes the flat value $\mu_w = 0.20$ ($k = 0$). The `stronge_realistic` row carries the video-calibrated curve $\mu_w =$

clamp$(0.20-0.4\,(v_{\text{slip}}-1),\ 0.20,\ 0.40)$: flat 0.20 for slip ≥ 1 m/s, rising toward 0.40 for slow, spin-dominated contacts. Both fitted Mathavan rows carry the same slope with re-fitted clamps, $\mu_w = \text{clamp}(0.20-0.4\,(v_{\text{slip}}-1),\ 0.16,\ 0.28)$. No published measurement of pool cushion $\mu_w(v_{\text{slip}})$ exists; the slopes are declared first-party calibration content (Section 3.2).

S3.5.1. Corrected Han impulse model (han2005_default)

As implemented at kernel `0.1.0+fb3425ad`, the default cushion model is the rigid-body impulse model of Han (2005), evaluated in a cushion frame with $\hat{\mathbf{x}}$ the horizontal normal from ball toward cushion, $\hat{\mathbf{y}}$ along the rail, and $\hat{\mathbf{z}}$ up. With (v_x, v_y, v_z) and $(\omega_x, \omega_y, \omega_z)$ the incoming components in that frame ($v_z \equiv 0$ for a bed-contact ball), the contact-point slip at the nose is (Han eqs. 14)

$$s_x = v_x \sin\theta_a - v_z \cos\theta_a + R\,\omega_y, \qquad s_y = -v_y - R\,\omega_z \cos\theta_a + R\,\omega_x \sin\theta_a, \tag{S27}$$

with slip speed $s_0 = \sqrt{s_x^2 + s_y^2}$, and the normal-channel scalar is $c = -v_x \cos\theta_a$. With the paper's coefficients $A = 7/2m$ and $B = 1/m$ (Han eqs. 16), the normal impulse and the slip-arrest impulse are (Han eqs. 17 and 20)

$$P_z^E = -\frac{(1+e)\,c}{B} = (1+e)\,m\,v_x \cos\theta_a, \qquad P_z^S = \frac{s_0}{A}, \tag{S28}$$

with $e = e(v_x)$ from Eq. (S23) and $\mu = \mu_w(s_0)$ from Eq. (S26). The case selection and tangential impulses are

$$(P_x^E, P_y^E) = \begin{cases} \big(s_x/A,\ s_y/A\big), & P_z^S \leq \mu\,P_z^E \quad \text{(Han eqs. 18)}, \\ \big(\mu P_z^E\, s_x/s_0,\ \mu P_z^E\, s_y/s_0\big), & \text{otherwise (Han eqs. 19)}, \end{cases} \tag{S29}$$

(the first case is sliding-then-adhesion, the second gross slip throughout the impact), transformed to the rail frame (Han eqs. 21–22) as $P_X = -P_x^E \sin\theta_a - P_z^E \cos\theta_a$, $P_Y = P_y^E$, $P_Z = P_x^E \cos\theta_a - P_z^E \sin\theta_a$, and applied as (Han eqs. 23)

$$\begin{aligned} v_x' &= v_x + P_X/m, & \omega_x' &= \omega_x - \tfrac{R}{I} P_Y \sin\theta_a, \\ v_y' &= v_y + P_Y/m, & \omega_y' &= \omega_y + \tfrac{R}{I}\,(P_X \sin\theta_a - P_Z \cos\theta_a), \\ v_z' &= v_z, & \omega_z' &= \omega_z + \tfrac{R}{I} P_Y \cos\theta_a. \end{aligned} \tag{S30}$$

Pocket-jaw arcs use identical physics with the normal taken from the arc center to the ball center.

Itemized deviations from the published paper (the content of "corrected" in Table 2):

1. **Slip/stick case-selection inequality.** As printed, the paper's boundary between the final-slip and final-stick cases omits the friction coefficient; the implemented test is $P_z^S \le \mu\, P_z^E$ as in Eq. (S29). Without the μ, near-normal impacts select the wrong branch. The correction follows the reference implementation (Kiefl, 2024), whose revision history documents it.
2. **Slip-direction angle θ_0.** The paper's eqs. 18–19 are written with $\sin\theta_0$, $\cos\theta_0$ for an angle that is never defined. The implementation fixes it to the initial contact-point slip direction, $\cos\theta_0 = s_x/s_0$, $\sin\theta_0 = s_y/s_0$, which is how it enters Eq. (S29) (same lineage; Kiefl, 2024).
3. **Restitution and friction inputs.** The paper's own speed-dependent restitution fit and incidence-dependent friction are not used; restitution and friction enter through the declared hooks of Eqs. (S23) and (S26) (constant $e = 0.85$, $\mu_w = 0.20$ in the frozen default row).

4. **Planar output.** The vertical impulse P_Z is computed (it feeds the ω_y update) but not applied to v_z; the vertical kinetic energy it would impart is treated as impact loss. This planar convention is shared with the reference implementation and with the Stronge model below.
5. **Contact geometry.** $\sin\theta_a = h/R - 1$ with the WPA nose height $h = 1.27R$, per Eq. (S22) — not the snooker $7R/5$ of Mathavan et al. (2010).

S3.5.2. Stronge tangential-compliance model (stronge_realistic)

As implemented at kernel `0.1.0+fb3425ad`, the compliant alternative is the planar rough-impact model of Stronge (2018, ch. 5): the contact point connects through independent normal and tangential springs, both phases are simple harmonic, and the impact is closed-form except for locating stick–slip transitions. The formulation reduces to a collinear problem in the initial tangential and normal contact-point velocities ($v_{t0} \le 0,\ v_{n0} < 0$), with v_{t0} measured against a unit tangent chosen opposite the slip.

Nondimensionalization. Only the frequency ratio is physical. Time is scaled so the normal contact frequency is $\omega_n = 1$; compression ends at $t_c = \pi/2$ and the impact at $t_f = (1 + e_n)\, t_c$, where $e_n = e(-v_{n0})$ from Eq. (S23), additionally clamped to $[0.01, 1]$ (a non-physical configuration must degrade, not overflow). The restitution-phase angle is $\phi(t) = t/e_n + \frac{\pi}{2}(1 - 1/e_n)$, and the normal channel closes with $v_{nf} = -e_n v_{n0}$. The tangential spring oscillates at $\omega_t = \omega_r$ (the *spin ratio* of Table 2: the tangential-to-normal contact-stiffness frequency ratio, equivalent to a cushion Poisson-ratio parameterization; in-domain $\omega_r \in (1, 2)$, frozen at 1.8), with the inverse-inertia coefficients of a

sphere on a half-space $\beta_t = 3.5$, $\beta_n = 1$, $\beta = \beta_t/\beta_n$, and $\eta^2 = \beta/\omega_r^2$. Friction is $\mu = \mu_w(|v_{t0}|)$ from Eq. (S26).

Branch structure. With $r = v_{t0}/v_{n0} \geq 0$:

$$\begin{aligned} &\text{initial stick:} && r < \mu\,\eta^2, \\ &\text{gross slip:} && r > \max\big(\mu\beta,\ \mu[(1+e_n)\beta - \eta^2/e_n]\big), \\ &\text{slip–stick–slip:} && \text{otherwise.} \end{aligned} \tag{S31}$$

The gross-slip branch closes in one line, $v_{tf} = v_{t0} - \mu\, v_{n0}\, \beta\,(1+e_n)$ (the Coulomb limit). For initial slip, the stick onset t_s has the closed form $t_s = t_c\, T$ with

$$T = \begin{cases} \dfrac{2}{\pi}\arccos\dfrac{(r/\mu)-\beta}{\eta^2-\beta}, & r \leq \mu\beta, \\[2ex] \dfrac{2e_n}{\pi}\left[\arccos\dfrac{(r/\mu)-\beta}{\eta^2/e_n - e_n\beta} - \dfrac{\pi}{2}\Big(1-\dfrac{1}{e_n}\Big)\right], & r > \mu\beta. \end{cases} \tag{S32}$$

(onset in compression and in restitution, respectively).

Stick evolution and the cone-crossing functions. During stick the tangential spring holds the contact (v_t and the spring displacement evolve harmonically at ω_t). For the *initial-stick* branch the spring loads from rest at $t = 0$, with displacement $u(t) = -(v_{t0}/\omega_t)\sin(\omega_t t)$; for the *slip–stick–slip* branch the harmonic stick episode starts at t_s from the state accumulated over the initial slip,

$$(u_0,\, v_0) = \begin{cases} \big(-\mu\, v_{n0}\, \eta^2 \sin t_s,\ \ v_{t0} - \mu\beta\, v_{n0}\,(1-\cos t_s)\big), & t_s \leq t_c, \\[2ex] \big(-\mu\, v_{n0}\, \eta^2 \sin\phi(t_s),\ \ v_{t0} - \mu\beta\, v_{n0}\,(1-e_n\cos\phi(t_s))\big), & t_s > t_c, \end{cases} \tag{S33}$$

with u_0 the spring displacement and v_0 the tangential contact velocity at onset. Slip resumes when the required spring force first crosses the Coulomb cone $\mu f_n(t)$, whose restitution-phase profile is proportional to $\sin\phi(t)$; the root-searched cone-violation functions are, for the initial-stick branch,

$$f_{\mathrm{IS}}(t) \;=\; \big|\, r\sin(\omega_t t)\,\big| \;-\; \mu\,\eta^2\,\omega_t\,\sin\phi(t), \tag{S34}$$

and for the slip–stick–slip branch, in terms of the normalized spring state $\alpha(t)$,

$$\alpha(t) = \frac{u_0\cos\big(\omega_t(t-t_s)\big) - (v_0/\omega_t)\sin\big(\omega_t(t-t_s)\big)}{\mu\, v_{n0}}, \tag{S35}$$

$$f_{\mathrm{SSS}}(t) \;=\; |\alpha(t)| \;-\; \eta^2\sin\phi(t), \tag{S36}$$

a cone violation being $f > 0$. At the terminal slip time t_{slip} (the first-crossing selection below) the exit velocity combines the harmonic stick state with the Coulomb friction accumulated over the remaining normal impulse: with $d = t_{\mathrm{slip}} - t_s$,

$$v_{tf}^{\mathrm{IS}} = v_{t0}\cos(\omega_t\, t_{\mathrm{slip}}) + \sigma\,\mu\, v_{n0}\,\beta\, e_n\,\big[1+\cos\phi(t_{\mathrm{slip}})\big], \tag{S37}$$

$$v_{tf}^{\mathrm{SSS}} = \omega_t\, u_0\sin(\omega_t d) + v_0\cos(\omega_t d) + \sigma\,\mu\, v_{n0}\,\beta\, e_n\,\big[1+\cos\phi(t_{\mathrm{slip}})\big], \tag{S38}$$

(the initial-stick and slip–stick–slip exits, respectively), where the sign $\sigma = \pm 1$ is set by the side of the spring at the crossing (friction on resumed slip opposes the slip the spring force drives): $\sigma = +1$ iff $-r\sin(\omega_t\, t_{\mathrm{slip}}) \geq 0$ for the initial-stick branch and iff $\alpha(t_{\mathrm{slip}}) \geq 0$ for the slip–stick–slip branch (in-domain, $e_n\omega_t > 1$, every first crossing lands on $\sigma = +1$). If no crossing exists, stick holds to t_f, where $\phi(t_f) = \pi$ and the friction term vanishes identically, leaving the pure harmonic value.

First-crossing selection rule. At stick onset the cone-crossing function has a structural double root ($f(t_s) = f'(t_s) = 0$ exactly: the spring rides the Coulomb limit at onset), so an unrestricted bracketing search can lock onto the tangency instead of the genuine crossing—flipping between root families at float-noise scale along an incidence sweep. This was the branch-selection defect (the 0.85-diamond bank-arrival discontinuity; case study in Section 6). The shipped selection rule (Section 3.5) scans the branch's cone-violation function—f_{IS} of Eq. (S34) or f_{SSS} of Eq. (S36)—over $K = 64$ equally spaced samples on $(t_{\mathrm{lo}}, t_f]$, with $t_{\mathrm{lo}} = t_c$ for the initial-stick branch and $t_{\mathrm{lo}} = \max(t_s, t_c)$ for the slip–stick–slip branch; the first sample with a positive cone violation closes a bracket containing a simple root, polished by a deterministic Newton-plus-bisection iteration with a fixed 64-iteration cap; if no sample is positive, stick holds to t_f. K is fixed, never adaptive. The pre-fix whole-bracket selection is retained only as an explicit legacy option for byte-exact replay.

Impulse application. The contact-velocity changes map to the ball through the inverse-inertia coefficients,

$$\Delta v_n = \frac{v_{nf} - v_{n0}}{\beta_n}, \qquad \Delta v_t = \frac{v_{tf} - v_{t0}}{\beta_t}, \qquad \Delta\boldsymbol{\omega} = \frac{5}{2R}\,(-\hat{\mathbf{n}}) \times \hat{\mathbf{t}}\;\Delta v_t, \tag{S39}$$

where $\hat{\mathbf{n}}$ is the three-dimensional contact normal tilted below the horizontal by θ_a from Eq. (S22) (a deviation from the reference implementation's $\tan\theta_a$ convention, worth about 0.5°), and the outgoing v_z is zeroed (planar convention, as in the Han model).

S3.5.3. Mathavan dual-contact model and the fitted presets (mathavan_fitted, grip_fitted)

As implemented at kernel `0.1.0+fb3425ad`, the fitted presets are built on the dual-contact impulse-space integration of Mathavan et al. (2010): the impulse accumulates in steps ΔP while Coulomb friction acts at *two* contact points, the cushion nose I (friction μ_w) and the table bed C (friction μ_s, the cloth sliding coefficient), the bed normal force growing as the nose pushes the ball down. In the cushion frame ($\hat{\mathbf{x}}$ along the rail, $\hat{\mathbf{y}}$ the horizontal normal into the cushion, approach $v_y > 0$), the contact-point slip velocities are

$$\begin{aligned} &\text{nose I:} \quad v_{xI} = v_x + \omega_y R \sin\theta_a - \omega_z R \cos\theta_a, \qquad v_{yI} = -v_y \sin\theta_a + \omega_x R, \\ &\text{bed C:} \quad v_{xC} = v_x - \omega_y R, \qquad\qquad\qquad\qquad\quad v_{yC} = v_y + \omega_x R, \end{aligned} \tag{S40}$$

with slip angles ϕ_I, ϕ_C and nose force ratios $(g_x, g_y) = \mu_w(\cos\phi_I, \sin\phi_I)$ on the pure-Coulomb path. One impulse step updates (the paper's eqs. 12–14, transcribed as implemented; $D = \max(\sin\theta_a + g_y \cos\theta_a,\, 0)$ is the bed normal-force factor)

$$\begin{aligned} \Delta v_x &= -\tfrac{1}{m}\big(g_x + \mu_s \cos\phi_C\, D\big)\Delta P, \\ \Delta v_y &= -\tfrac{1}{m}\big(\cos\theta_a - g_y \sin\theta_a + \mu_s \sin\phi_C\, D\big)\Delta P, \\ \Delta \omega_x &= -\tfrac{5}{2mR}\big(g_y + \mu_s \sin\phi_C\, D\big)\Delta P, \\ \Delta \omega_y &= -\tfrac{5}{2mR}\big(g_x \sin\theta_a - \mu_s \cos\phi_C\, D\big)\Delta P, \\ \Delta \omega_z &= +\tfrac{5}{2mR}\, g_x \cos\theta_a\, \Delta P. \end{aligned} \tag{S41}$$

Compression integrates until $v_y \le 0$ (the crossing re-approached with eight fixed step-halvings), accumulating the nose normal work W_{comp} via $\mathrm{d}W = \Delta P\,|v_y|\cos\theta_a$; restitution integrates until the nose work reaches the energetic

target

$$W_{\text{rest}} \;=\; ee^2\, W_{\text{comp}}, \tag{S42}$$

with $ee = ee(v_{y0})$ from Eq. (S24) and the final step sized exactly to the remaining work. The step is $\Delta P = \max(m\, v_{y0}/N,\ 10^{-4}\,\text{kg m/s})$ with $N = 1000$ (restitution analogously from the target work), a hard cap of $10N$ steps per phase raising a diagnostic error (never silent truncation), and impacts whose normal approach exceeds the 2.5 m/s experimental validity bound of the source are counted per shot rather than rejected. Itemized deviations from the published paper: (i) restitution enters through the energetic hook of Eq. (S24) rather than the paper's fixed snooker fit; (ii) nose friction enters through Eq. (S26) at the arrival nose-slip speed, bed friction is the cloth μ_s; (iii) the bed normal-force factor is clamped at zero ($D \geq 0$: the unilateral bed cannot pull); (iv) the paper's §3.2 rolling-condition force nulling (its eqs. 15a/15b) is omitted—measured effect below +0.06 on the rolling-perpendicular exit-speed ratio—because an epsilon-free rolling detector does not exist in impulse stepping; (v) the incoming v_z passes through unchanged (planar convention).

Tangential compliance (both fitted presets). The fitted presets add a first-party tangential-compliance element at the nose: a Cundall–Strack-style bristle spring per nose tangent direction, fed by the contact-point slip, its per-step force Coulomb-capped at $\mu_w N$. The impulse-to-time map is a half-sine normal-force profile, floored and argument-capped, over the a-priori total-

impulse estimate:

$$N(p) = \max\left[N_{\text{peak}} \sin\!\Big(\frac{\pi\,\bar{p}}{P_{\text{tot}}}\Big),\; N_{\text{peak}} \sin(0.02\,\pi)\right], \quad \bar{p} = \min(p,\, 0.999\,P_{\text{tot}}), \tag{S43}$$

$$N_{\text{peak}} = \frac{\pi\,P_{\text{tot}}}{2\,\tau_c}, \quad P_{\text{tot}} = \frac{m\,v_{y0}\,(1+ee)}{\cos\theta_a}, \quad k_t = \Big(\frac{\chi\pi}{\tau_c}\Big)^2 \frac{m}{\beta_t}, \tag{S44}$$

where p is the accumulated normal impulse and the integrator advances internal time as $\mathrm{d}t = \Delta P/N(p)$. The $\sin(0.02\pi)$ floor and the $0.999\,P_{\text{tot}}$ argument cap are part of the frozen arithmetic, not implementation detail: the bare profile gives $N(0) = 0$, making the first step's $\mathrm{d}t$ singular, and both guards reshape the profile tails. The internal time scale τ_c cancels identically (only the frequency ratio χ is physical, the same invariance as the Stronge normalization) and $\beta_t = 3.5$. The peak elastic energy stored in the follow-channel spring is held as a reservoir and returned at release as an impulse along the stored displacement with gain e_x (energy returned $= e_x^2 \times$ held), capped by both the whole-impact kinetic-energy deficit and the preset-dependent Coulomb allowance $\mu_{\text{rel}}\,(P_{\text{tot}} - P_{\text{comp}})$, with P_{comp} the normal impulse accumulated at the end of compression. The release ceiling is the *static* grip bound for a captured `grip_fitted` impact—$\mu_{\text{rel}} = \mu_{\text{grip}} = 2.5\,\mu_w$ (the held reservoir is grip-phase elastic storage; its give-back flows through the gripped interface, not a sliding one)—and the kinetic $\mu_{\text{rel}} = \mu_w$ otherwise (`mathavan_fitted` and uncaptured arrivals); e_x is a phenomenological give-back gain, not a Cross-convention tangential restitution. `mathavan_fitted` freezes $\chi = 0.8$, $e_x = 0.9$ (jointly fitted with $ee_0 = 0.80$ and the μ_w clamps above against a high-speed-video response surface (Alciatore, n.d.b), with the source's own rebound figures held as anchors). `grip_fitted` additionally enables a static/kinetic stick–slip mode per nose channel: the loaded

spring transmits force up to the static ceiling $\mu_{\text{grip}} = 2.5\,\mu_w$ while gripping, engaged only for spin-overdriven arrivals with nose slip at or below 0.6 m/s (capture threshold), with channel disengage when the spring unloads through its natural length and a whole-impact energy ledger that scales tangential exit degrees of freedom down if the loop would otherwise exit above the arrival kinetic energy. Its constants are $\chi = 1.2$, $e_x = 0.9$, grip ratio 2.5, capture 0.6 m/s, paired with Eq. (S25). Both presets' declared validity domains and out-of-domain observables are ledgered; neither was re-tuned.

S3.6. Compliant frozen-cluster resolution (CompliantChain)

As implemented at kernel `0.1.0+fb3425ad`, an impact reaching a frozen cluster (any chain of contacts within 10^{-5} m surface separation, swept transitively) escalates to a bounded compliant episode (Section 3.3). Every active contact carries the Hertz–Lankarani–Nikravesh normal force (Lankarani and Nikravesh, 1990, 1994)

$$F_N \;=\; \delta^{3/2}\left(K + \chi_{\text{LN}}\,\dot{\delta}\right) \;\geq\; 0, \qquad \chi_{\text{LN}} \;=\; \frac{3K\,(1-e_b^2)}{4\,\max(\dot{\delta}^{-},\,10^{-3}\,\text{m/s})}, \tag{S45}$$

with δ the penetration depth, e_b the mean ball–ball restitution of the pair, and $\dot{\delta}^{-}$ the contact's impact speed, maintained as the largest *equivalent* impact speed $\sqrt{\dot{\delta}^2 + 4K\delta^{5/2}/(5m_{\text{eff}})}$ the contact has exhibited (the Hertz energy inversion; for clean impacts this equals the first-penetration approach speed exactly, and for contacts entering a window mid-compression it restores the Lankarani–Nikravesh definition of $\dot{\delta}^{-}$ as the impact velocity). The force is clamped non-negative (cohesionless contact), and the grazing clamp bounds the damping singularity as $\dot{\delta}^{-} \to 0$. The Hertz stiffness is derived, not fitted:

$$K = \tfrac{4}{3}\,E^*\sqrt{R^*}, \qquad \frac{1}{E^*} = \frac{2\,(1-\nu^2)}{E}, \qquad R^* = \frac{R}{2}, \tag{S46}$$

with $E = 3800\,\mathrm{MPa}$ and $\nu = 0.375$ for phenolic resin (Gismeros Moreno et al., 2022, Table 1) and the catalog ball radius, giving $K = 3.524 \times 10^8\,\mathrm{N/m^{3/2}}$. Tangential traction is force-based Coulomb, $\mu_{bb}(v_{\mathrm{slip}})\,F_N$ opposing the contact-point slip with the same friction curve as Eq. (S11) (no impulse cap is consumed—the property that kept the compliant rows bit-stable across the mid-program re-baseline, Section 3.3).

All contacts advance *concurrently* (Jacobi accumulation: every force reads only previous-step state, making the episode order-free and mirror-equivariant) under semi-implicit Euler with fixed step $\mathrm{d}t = 10^{-6}\,\mathrm{s}$ and a hard cap of 3000 steps (a 3 ms window bounding the 1–2 ms break transient). The episode ends early when every contact is open ($\delta \le 0$) and non-approaching (normal closing speed below $10^{-9}\,\mathrm{m/s}$), or hands off mid-window when a non-member pair first closes within $10^{-5}\,\mathrm{m}$ while approaching (the event loop detects the new contact and re-escalates with the newcomer as a member). Three post-window guards fail closed, restoring the pre-window state and running the legacy resolver on the same inputs: non-finite state; cluster energy gain (kinetic plus stored Hertz elastic energy $\frac{2}{5}K\delta^{5/2}$ exceeding the pre-window value by more than $10^{-9}\,\mathrm{J}$); and cap expiry with a still-penetrating approaching pair. The `SequentialImpulse` legacy arm resolves the same cluster by deterministic Gauss–Seidel sweeps of pairwise frictional impulses (at most 256 sweeps) and is retained as an explicit recorded option; it is the baseline class whose starved momentum exchange produces the dead-cue break defect that experiment E3 gates on (Section 3.3).

S3.7. Event detection, epsilons, and rest projection

As implemented at kernel `0.1.0+fb3425ad`, every candidate event time is a real root of a polynomial in time of degree at most four, because every ball-position component is at most quadratic (S3.1)—with one *scheduled* exception, the airborne pocket heuristic below. Per event class: ball–ball proximity $|\Delta\mathbf{r}(\tau)|^2 = (2R)^2$ between two quadratic paths is quartic; a linear cushion (vertical plane at signed distance R) is quadratic; jaw arcs and pockets (vertical cylinders, contact at arc radius $+R$; capture of a *grounded* ball when the *center* crosses the pocket radius) are quartic; the regime transitions have the closed forms τ_{slide}, τ_{roll}, τ_{spin} of S3.1 and need no root-finding. Quartics are solved with the closed-form solver of Orellana and De Michele (2020) incorporating both published corrections (De Michele, 2022, 2024), plus two documented numerical accommodations inherited from Kiefl (2024): the near-degenerate-quadratic threshold in the LDLT factorization is loosened by a factor of 100 so that Newton's-cradle-like quartics (nearly equal velocities) yield four distinct roots rather than spurious doubles, and a root is accepted as real under a hybrid filter—$|\text{Im}| < 10^{-9}$ when $|\text{Re}| > 10^{-3}$, else $|\text{Im}|/|\text{Re}| < 10^{-3}$. The earliest accepted root strictly beyond the 10^{-9} s simultaneity guard becomes the next event; a leading coefficient is degraded (quartic to cubic, and so on) when it cannot exceed one machine epsilon of the next term anywhere within the Cauchy root bound. Contact at $\tau = 0$ fires immediately only for approaching pairs; an at-contact cushion root requires approach speed above 10^{-5} m/s (the sleeping threshold that lets resting contact coexist with detection).

Airborne pocket capture (non-root scheduling). For an *airborne* ball, pocket capture is not a polynomial root: it is a documented phenomenological approximation inherited from Kiefl (2024). With τ_{land} the ballistic descending crossing of $z = R$: (i) if the landing point lies strictly inside the pocket radius, capture is scheduled at $\tau_{\text{land}} - \epsilon_t$ (the 10^{-9} s time epsilon), guaranteeing temporal precedence over the bed-landing event; (ii) otherwise the horizontal flight path (linear in xy) is intersected with the pocket cylinder, giving quadratic entry/exit roots $\tau_{\text{in}} < \tau_{\text{out}}$, and capture is scheduled at the midpoint $(\tau_{\text{in}} + \tau_{\text{out}})/2$ iff $\tau_{\text{in}} \geq 0$, $z(\tau_{\text{in}}) \geq R$ (the ball did not land first), and $z(\tau_{\text{out}}) \leq \frac{7}{5}R$ (the rim heuristic: low fliers ricochet off the back of the pocket and drop in, higher ones fly over); otherwise no airborne capture candidate exists and the landing proceeds. A jump-shot reproduction that treats pocket capture as a root of the quartic above will mis-order these events.

Frozen tolerance table: time, space, velocity, and spin epsilons all 10^{-9} (s, m, m/s, rad/s); root-acceptance bounds as above; separation ("kiss") gap 10^{-6} m; frozen-rack surface gap 0.5 μm (below the 10^{-5} m cluster threshold, so racked contacts are frozen topology; seeded back-row micro-gaps 0.05–0.5 mm sit above it).

Frozen-rack topology (seeded selection rule). The default rack constructor realizes the frozen topology with frozen fraction $f = 0.7$ (a declared first-party consensus estimate; no published frozen-fraction statistics exist). The apex wedge—apex plus row 1—always keeps its lattice nodes and consumes no random draw. Each back-row ball (lattice row ≥ 2), visited in rack order against the shot-seeded deterministic stream, keeps its node with probability f; otherwise it is displaced by a micro-gap $g \sim U[0.05, 0.5)$ mm along a

direction $\vartheta \sim U[0, 2\pi)$, clamped so no pair drops below the frozen lattice pitch. Then each back-row boundary $r{-}1 \to r$ ($r \geq 2$, ascending, same stream) opens with probability $1 - f$ by shifting all rows $\geq r$ away from the breaker by an independent g. The same seed therefore regenerates the exact rack topology of experiment E3 bit-for-bit; the fraction and rule are required inputs, not tuning detail.

Terminal rest projection (round-trip law). Every simulation ends by projecting all balls to exact rest, and any pair the input validator would reject—center distance below $r_a + r_b - 10^{-6}$ m—is retracted along the line of centers, each ball moving half the correction, to the exact 10^{-6} m gap, sweeping pairs in ball-identity order to a fixed point (at most 128 passes; observed convergence in single-digit passes), then re-seated to $z = R$ and clamped against rail-seat bands and pocket capture discs. Pairs the validator accepts, including undisturbed frozen-rack contacts, are never touched, so clean records remain byte-identical. The result is the round-trip law of Section 3.5: every `simulate` output is a valid `simulate` input, with no accumulating representational debt over arbitrarily long shot chains.

S3.8. Consolidated frozen constants

Table S2 consolidates every constant fixed by the frozen catalog, each with units and provenance class: *lit* (published value or derived from published values), *fit* (first-party calibration, declared as such), or *num* (numerical guard or discretization constant). All values are as implemented at kernel `0.1.0+fb3425ad` and were frozen before any battery sweep ran (Section 4).

Table S2: Consolidated frozen constants of the validated models. Provenance classes: lit = published/derived, fit = first-party calibration (declared), num = numerical guard. Equation references are to this appendix.

Constant	Value	Class	Provenance / role
Ball, cloth, and gravity (S3.1)			
Ball mass m	0.170097 kg	lit	regulation set (Kiefl, 2024)
Ball radius R	0.028575 m	lit	57.15 mm US ball
Gravity g	9.81 m/s^2	lit	—
Sliding friction μ_s	0.20	lit	worsted tournament cloth (Alciatore, n.d.a)
Rolling friction μ_r	0.010	lit	lag-speed calibration class (Section 5)
Spin-decay μ_{sp}; rate α	$\frac{4}{9}(R/\mathrm{m}) = 0.0127$ (dimensionless); $\frac{10}{9}g \approx 10.9$ rad/s^2	lit	video-measured ≈ 10 rad/s^2 (Alciatore, n.d.c); Eq. (S6)
Ball–slate restitution e_t	0.5	lit	reference default (Kiefl, 2024); no published measurement; Eq. (S7)
Slate bounce guard (apex)	5 mm (default) / 10 μm (realism presets)	num	Zeno guard, S3.2
Ball–ball impact (S3.3)			

Constant	Value	Class	Provenance / role
Restitution e_b	0.95	lit	community constant: Alciatore (n.d.a) physical-properties compilation (band 0.92–0.98, printed without attribution; no first-hand measurement located; the $[0.92, 0.98]$ bracket is a declared sensitivity range, not a measured interval — main-text Section 4); ledgered as declared model content
$\mu_{bb}(v)$: (a, b, c)	(0.009951, 0.108, 1.088)	lit	Alciatore (n.d.d) fit; Eq. (S11)
Slip-impulse cap	$(1 + e_b)/2 \cdot \lvert\Delta v_n^-\rvert$	lit	TP A.5 Eq. 10; corrected default (slip-law fix)
Legacy cap (replay only)	$e_b \lvert\Delta v_n^-\rvert$	lit	pre-correction law (Kiefl, 2024)
Micro-hop keep threshold	5 mm (default) / 10 μm (realism presets)	num	bounce-apex guard, S3.3
Stick–ball strike (S3.4)			
Cue mass M	0.567 kg	lit	20 oz cue

Constant	Value	Class	Provenance / role
Tip radius ρ	0.0106 m (nickel); 0.00895 m dime variant	lit	TP B.1 curvature standards
Default endmass m_e	$m/30 \approx 5.67$ g	lit	TP A-31 (Alciatore, n.d.e)
Endmass catalog span	1.70–12.7 g	lit	pivot-length inversion $m/m_e = 2.5L_p/R - 3.5$ (Shepard, 2001); robot-measured L_p
Tip restitution e_{tip}	1.0 default; 0.73 leather; 0.85 phenolic	lit	measured tip efficiency (Alciatore, n.d.b)
Miscue envelope	$a_{\text{cmd}}^2 + b_{\text{cmd}}^2 \leq 0.25$	lit	tip-offset limit, Eq. (S15); contact-offset binding post-registration (S3.4)
Side-offset sign map	$a = -a_{\text{cmd}}/(1 + \rho/R)$	num	commanded (right-English positive) $\rightarrow$ internal (left positive); Eq. (S16)
Shot speed domain	$v_0 \in (0, 15]$ m/s	num	input validity
Pinch onset / full	60° / 70°	lit	jump-vs-massé regime boundary (Coriolis, 1835); Eq. (S20)
Press return κ; relax τ_{pinch}	0.45; 0.12 s	fit	massé footage fit; Eq. (S21)
Cushion geometry and hooks (S3.5)			

Constant	Value	Class	Provenance / role
Nose height h (US pool)	$1.27R$ ($\sin\theta_a = 0.27$)	lit	WPA specification; Eq. (S22)
Nose height h (snooker pairing)	$1.4R$ ($\sin\theta_a = 0.4$)	lit	Mathavan et al. (2010)
Kinematic e_0; clamp	0.85; $[0.6, 0.98]$	lit	Han (2005) lineage; Eq. (S23)
Energetic ee_0 (default / `mathavan_fitted`)	0.98 / 0.80	lit / fit	snooker fit (Mathavan et al., 2010) / joint video fit
`grip_fitted` $ee(v)$	$0.65 + 0.18(v_n - 1)$, clamp $[0.5, 0.98]$	fit	Eq. (S25)
μ_w (`han2005_default`)	0.20 flat	lit	no published $\mu_w(v)$ exists (declared)
μ_w (`stronge_realistic`)	$0.20 - 0.4(v - 1)$, clamp $[0.20, 0.40]$	fit	video kick/sidespin anchors; Eq. (S26)
μ_w (fitted presets)	$0.20 - 0.4(v - 1)$, clamp $[0.16, 0.28]$	fit	joint fit with tangential compliance
Stronge model (S3.5.2)			
Spin ratio ω_r	1.8; domain $(1, 2)$	lit	Stronge (2018) parameterization; reference default (Kiefl, 2024)
Inverse inertia β_t, β_n	3.5, 1	lit	sphere on half-space (Stronge, 2018)
e_n clamp	$[0.01, 1]$	num	degradation guard
First-crossing scan K; root cap	64; 64 iterations	num	fixed-resolution selection (branch-selection fix)

Constant	Value	Class	Provenance / role
Mathavan model and fitted presets (S3.5.3)			
Impulse step ΔP	$\max(mv_{y0}/1000, 10^{-4})$ kg m/s	num	first-order bias control
Step caps; refinement	10×1000 per phase; 8 halvings	num	diagnostic overrun, never silent
Validity bound	$v_n \leq 2.5$ m/s (counted)	lit	source experimental range (Mathavan et al., 2010)
Bristle force floor; cap	$N \geq N_{\text{peak}} \sin(0.02\pi)$; $p \leq 0.999\, P_{\text{tot}}$	num	profile-tail guards; Eq. (S43)
`mathavan_fitted` χ, e_x	0.8, 0.9	fit	joint HSV-response fit; Eq. (S44)
`grip_fitted` χ, e_x, grip ratio, capture	1.2, 0.9, 2.5, 0.6 m/s	fit	grip-mode joint fit
Break cluster (S3.6)			
Hertz stiffness K	3.524×10^8 N/m$^{3/2}$	lit	derived, Eq. (S46) (Gismeros Moreno et al., 2022)
Step; cap	10^{-6} s; 3000 steps (3 ms)	num	Hertz contact-time resolution
Grazing clamp on $\dot{\delta}^-$	10^{-3} m/s	num	Lankarani–Nikravesh singularity guard; Eq. (S45)
Cluster contact threshold	10^{-5} m	num	escalation / hand-off tolerance
Energy-audit slack	10^{-9} J	num	fail-closed guard

Constant	Value	Class	Provenance / role
Legacy sweep cap	256	num	`SequentialImpulse` bound
Event detection and rest (S3.7)			
Time/space/velocity/spin eps	10^{-9} (s, m, m/s, rad/s)	num	simultaneity and zero guards
Root-real filter	$\lvert\mathrm{Im}\rvert < 10^{-9}$ if $\lvert\mathrm{Re}\rvert > 10^{-3}$, else ratio $< 10^{-3}$	num	hybrid acceptance
Degeneracy relaxation	$\times 100$	num	cradle-quartic guard (Kiefl, 2024)
Separation gap	10^{-6} m	num	retraction target; round-trip law
Frozen-rack gap; seeded gaps	0.5 μm; 0.05–0.5 mm	num	frozen-contact topology (Section 3.3)
Frozen fraction f (rack)	0.7; apex wedge always frozen, back rows sampled	fit	racking-consensus estimate (declared); selection rule in S3.7
Rest sleeping threshold	10^{-5} m/s	num	at-contact approach gate
Rest projection pass cap	128	num	fixed-point sweep bound

S4. Extended results: groups B and D

This section carries the per-experiment result prose behind the group summary tables of the main text's Sections 5.2 (stick–ball) and 5.4 (curved paths). The figures cited here (Figs. 7 and 11) and the miscue-convention table (Table 8) appear in the main document.

S4.1. Group B — stick–ball, per experiment

Endmass squirt (B1). The kernel's cue-ball deflection (squirt) is the endmass closed form of TP A-31 (Alciatore, n.d.e) and Cross (2008), in the lineage of Shepard (2001). Over contact offsets 0–0.5R, end masses $\{5, 10, 15\}$ g, and speeds $\{1, 2, 4\}$ m/s plus the six-cue catalog grid (333 runs), the maximum comparison error against the closed form is 8×10^{-15} degrees — machine precision, confirming the implementation and the tip-offset plumbing end to end (Fig. 7a). The source's printed example evaluations reproduce to $|E| \leq 4.3 \times 10^{-4}$ degrees, the PDF's own rounding scale. The registered qualitative features quantify as: near-linearity in offset with $R^2 = 0.9967$–0.9979 per registered end-mass/speed series ($R^2 \geq 0.9961$ for the plotted six-cue catalog grids of Fig. 7a), the local slope density falling ∼12% toward the miscue limit (mildly sub-linear); speed invariance exact (measured spread 0.0° — the closed form contains no speed); and left/right symmetry to 1.4×10^{-14} degrees.

Natural pivot length (B2). A simulated aim-and-pivot binary search per cue recovers the natural pivot length L^* at which backhand-english pivoting self-corrects squirt; geometric consistency with the closed form holds to 4.8×10^{-10} m, the bisection floor. TP B.26's claim that one pivot length serves all offsets is strongly confirmed: re-using each cue's single L^* across the legal offset ladder leaves a worst-case residual aiming error of 0.077° (shipped catalog cues 0.006–0.026°). The comparison's boundary is a convention (Fig. 7b): catalog pivot lengths span 10.5–64.9 in, with only the break cue inside the Platinum-protocol 8–14 in band (the next-lowest shaft misses the band top by 0.03 in), and the companion squirt-angle comparison of B1 shows why —

commanding the protocol's published 6/12 mm as *tip-center* offsets (the protocol page's literal wording) undershoots all four robot-measured shafts by 0.32–0.42°, while reading the same numbers as *contact* offsets closes the gap almost exactly (+0.02° on the lowest-squirt shaft) with a growing positive residual toward the break cue (+0.22°). Re-referencing the pivot geometry to contact offsets scales L^* by $1/(1 + r_{\text{tip}}/R) \approx 0.729$, which would bring two further shafts inside the band. Both readings are recorded and neither is adopted as a recalibration: the commanded/tip-center form is the physical backhand-english geometry. In u_{val} terms (Table 4) the robot source publishes a band with no per-shaft uncertainty, and the 0.32–0.42° convention gap is an order larger than any plausible protocol u_D, so the boundary here is convention, not measurement uncertainty.

Miscue limit and maximum sidespin (B3). The registered "near-linear" spin-versus-offset question resolves to exactly linear: the source relation is linear in contact offset — spin-rate factor SRF $= \omega R/v = (5/2)(x/R)$ — and the kernel reproduces it to $|E| \leq 2.2 \times 10^{-16}$ in SRF with fitted quadratic curvature $\leq 5.5 \times 10^{-15}$, exact odd symmetry, and exact speed invariance (252 runs, plus a six-run dime-tip arm). The envelope, by contrast, is a conventions result (Table 8; Fig. 7c). The kernel's miscue rule is input validation on *commanded tip-center* offsets, $a^2 + b^2 \leq (0.5R)^2$, with no tip-friction model anywhere; the source (TP 2.1) derives the limit from tip friction at the *contact* offset. Under the shipped nickel tip the reachable contact offset therefore tops out at $0.365R$ (SRF = 0.912, below natural-roll-rate sidespin), and the tip friction coefficients implied by the kernel's ceilings (0.39–0.58 across conventions) sit at or below the 0.577–0.659 the literature attributes

to chalked tips — every accepted command is physically non-miscuing, so the conservatism is one-sided. Contact offsets in $(0.365, 0.5]R$ are physically legal per the source yet unreachable under the nickel tip, and $(0.5, 0.55]R$ is unreachable under every convention. The registered 0.525–0.6R sweep tail was realized as rejection probes: all 16 cells reject with identical diagnostics under both tip radii — designed rejection evidence for the envelope, not a failure mode. Post-registration, a kernel amendment (kernel `0.1.0+8efd660e`) re-binds the miscue limit to the contact offset (commanded cap $0.5\,(1 + r_{\text{tip}}/R) \approx 0.686R$, nickel); every previously accepted input reproduces bit-exactly. The rejection probes above are boundary probes of the registered envelope itself: under the revised, cue-aware convention the eight nickel-tip cells (contact 0.383–0.438R, inside the 0.5R contact limit) and the registered combined-axis probe at commanded $(0.375, 0.375)$ would be accepted and simulate, the eight contact-mode cells still reject, and the recorded tip-radius-independent diagnostics are superseded by cue-aware validation by design. The pinned artifact stands unchanged as the registered-envelope record; the post-registration maintenance re-baseline under the revised envelope, which measures exactly this disclosed class, is recorded in the study's baseline manifest. $(0.5, 0.55]R$ remains unreachable (amendment note: main-text Section 8).

Center of percussion (B4). Coriolis' center-of-percussion result — immediate natural roll when the tip strikes at 70% of ball height, i.e., a contact offset of 0.4R above center (Coriolis, 1835; Alciatore, 2008) — verifies end to end through the event pipeline over 41 offsets × three speeds (123 runs; Fig. 7d). The slide-phase duration is exactly zero at $b = 0.4R$ at all three speeds;

the interpolated crossing sits at $b^* = 0.4R + 1.4 \times 10^{-11}$ (the documented contact-mode tip correction), equivalently at height fraction 0.700, and the slide-time V-shape's arms match the closed form. The spin-rate factor follows $(5/2)(b/R)$ exactly (fitted slope 2.5, $R^2 = 1$). Above the crossing the cue ball overspins and cloth friction acts *forward*: the roll-speed conversion $v_{\text{roll}} = v(5+2\,\text{SRF})/7$ holds to machine precision, giving +7.14% speed at $b = 0.5R$; below it, a draw strike retains 35.7% of strike speed at $b = -0.5R$ and never reverses direction without an object ball. The slide map matches the closed forms to 6.3×10^{-11} m in slide distance and 1.8×10^{-11} s in duration over the 104 uncensored cells; the 19 censored cells are all $v_0 = 3$ m/s near-stun or draw shots whose closed-form slide distance (up to ≈3.2 m) exceeds the available ∼2.44 m runway — slides that cannot complete on a regulation 9-ft table either. They are censored by the foot cushion, characterized via the closed form, and the design was never re-aimed (a registered exclusion, not a waiver).

S4.2. Group D — curved paths (Coriolis), per experiment

Parabolic slide path (D1). Experiment D1 verifies that the cue ball's sliding path after an oblique rolling-ball impact is parabolic: half-ball hits at 1, 2, and 3 m/s, both cut signs, sampled at 1 ms. The registered acceptance criterion — quadratic-fit $R^2 > 1 - 10^{-6}$ in the frame aligned with the measured post-impact velocity — fails on all six runs, with $1 - R^2$ between 2.2×10^{-6} and 3.7×10^{-6}. The miss is a frame artifact, and we keep it as this paper's worked example of a frame-sensitive verification criterion: the real-ball impact model ($e_b = 0.95$, finite ball–ball friction) leaves the friction acceleration 2.9–4.1° off perpendicular to that axis, injecting a cubic term that a

quadratic fit cannot absorb. In the slip-aligned frame — the parabola's own axis frame — the fit is exact to machine precision: $1 - R^2 = 0.0$ on every run, worst time-domain residual 3.6×10^{-16} m, and the TP A.4 Eq. 20 slide duration reproduced to within 2.8×10^{-17} s (Alciatore, n.d.c) (Fig. 11a). The path extent scales as $v^{2.09}$–$v^{2.14}$ across the speed grid against the registered $\sim v^2$ claim, and the final direction is nearly speed-invariant: the comparison error against the TP A.4 closed-form angle anchor is $+0.17°$ at 1 m/s (the post-cascade value) growing to $-0.94°$ at 3 m/s, a $1.11°$ spread over the grid.

Massé aiming (D2). Experiment D2 tests the massé BAR aiming rule of TP A.19 (Alciatore, n.d.c) — "B for ball contact point, A for cloth aim point, R for CB resting point," the source's own naming (Alciatore, n.d.a): the final cue-ball direction should parallel the line from the resting point (R) through the aim point (A) on the cloth. Over elevations 45–75° (a declared executed subset of the registered 45–80° range), eight azimuths, and two tip speeds (64 cells; two ill-conditioned cells flagged and excluded from the headline statistics), the final-heading error has mean 0.68° and maximum 1.60°, improving with elevation to a maximum of 0.49° at the grounded 75° full-pinch massé (the 45–55° cells run in the airborne jump-launch regime). Two properties stand out. First, Coriolis' invariance theorem — friction acting along the fixed slip direction cannot rotate the final direction (Coriolis, 1835) — is realized exactly: the per-cell heading errors at the two tip speeds are identical to the full recorded precision of the output in all 32 (elevation, azimuth) pairs, so the per-speed error statistics coincide bit-for-bit (Fig. 11b). Second, 88% of cells "come up a little short" of the predicted direction, which is precisely the behavior the source itself expects of real shots (TP A.19, p. 3)

— a pass with the source's own caveat reproduced. The rule is also usable where it is taught: the final direction is realized within 0.75 m at worst (median 0.18 m), never beyond half a table length.

A scope note on the sub-pinch rows. The 45–55° cells run in the airborne jump-launch regime (the elevated strike launches the ball off the cloth), and there everything *after* the strike is inert to the comparison by a conservation identity: under any impulsive strike the natural-roll final heading — the azimuth of the table-plane combination $\mathbf{w}_{xy} = (5\mathbf{v}_{0,xy} + 2R\,\boldsymbol{\omega}_0 \times \hat{\mathbf{z}})/7$, the $\times\hat{\mathbf{z}}$ term planar by construction — is exactly invariant to intervening hops, to the ball–slate restitution, and to the sliding-friction law: every cloth-contact friction impulse enters $\mathbf{v}_{xy}$ and $R\,\boldsymbol{\omega} \times \hat{\mathbf{z}}$ with equal and opposite weights in that combination, and a normal (restitution) impulse at the cloth contact carries no planar component and no torque about the center. Coriolis makes the robustness claim himself — the constructions giving the final directions "subsistent toujours, quelles que soient les lois que suivent les deux frottemens de glissemens et de roulement" (quoted in Bouleau, 2006) — and the property is proven independently for this kernel's strike–flight–contact pipeline (a launch-invariance theorem in the kernel's unpublished design record). The 45–55° rows therefore exercise the strike's impulse–momentum mapping together with the flight and contact solver's conservation structure — a strike defect that mis-mapped tip offset to launch velocity or spin would still miss the R→A line, so the rows are not vacuous — but within the impulse–momentum model class the BAR construction is reproduced by derivation, so agreement there cannot discriminate strike *realism* beyond that idealization (finite tip dwell, tip–slate jamming) at those elevations; we also know of

no published massé-aiming measurement in the 30–60° elevation band. The flight-leg half of the claim is directly falsifiable in-sim: perturbing the ball–slate restitution e_t at 45° elevation must leave each cell's BAR heading error unchanged to output precision — any larger shift would indicate a broken flight/contact solver, not new strike physics.

Swerve maps (D3). Experiment D3 maps swerve over elevations $\{0, 5, 10, 20, 30\}°$, english fractions $\{25, 50, 100\%\}$, and tip speeds 1–3 m/s, with drift read at 0.5–2.5 m along the aim line (TP B.23 provenance; the registered claims are the FAQ's structural rules, Alciatore, n.d.c,n). All three registered structural rules hold with zero violations on the grid: no drift at level cue (the maximum |curve| over all level-cue rows is exactly zero — pure vertical-axis spin leaves the sliding leg straight), drift monotone in elevation and in spin, and curvature at fixed distance decreasing with speed, the delayed-onset rule. The delayed onset is made quantitative through the squirt–swerve cancellation distance ("squerve"): the distance at which swerve re-crosses the aim line that the 1.9° full-english squirt displaced. Measured cancellation distances span 0.08–2.65 m on the grid; at 5° elevation and full english the distance moves outward from 0.26 m at 1 m/s to 1.05 m at 2 m/s (the marked 5° crossing in Fig. 11c) to 2.36 m at 3 m/s — a derived, teachable quantity linking the stick–ball and curved-path channels. Cells in which the ball rests, or the 3.2 m observation window ends, before re-crossing record no distance. The sliding legs are closed-form parabolas to within 2.6×10^{-13} m — the same verification floor as D1 — and the experiment is cue-only, bit-identical across the slip-law kernel cascade.

S5. Predicted-shift verification across the slip-law and branch-selection fix cascade

Before the post-fix battery re-run, each affected experiment received a written prediction of the direction and mechanism of its shift, and no registered band or catalog parameter was re-tuned (Section 6 of the main text presents the fix-cascade case study and a two-row extract). Table S3 transcribes all seven prediction classes; all seven confirmed.

Table S3: Predicted-shift verification across the slip-law and branch-selection fix cascade. Each row is a prediction class registered before the post-fix battery re-run; "bit-identical" means byte equality of the affected rows after masking kernel-version stamps. No acceptance band or catalog parameter was re-tuned.

Expt.	Registered prediction	Measured outcome
C1	branch-selection fix: the four `stronge_realistic` discontinuous bank cells collapse; other presets untouched	cells $4 \to 0$; the three-crossing cell (aim 3.0, 1 m/s) monotone; `han2005`, `mathavan`, and `grip` rows bit-identical
A3	slip tan-argument ratio $\to 1$ (throw exactly $\arctan \mu$); stick regime unchanged	ratio $0.9745 \to 1.00014$; stick ratio 1.02564 unchanged; worst slip-channel $\lvert E \rvert$ vs the transcribed master curves $0.0886 \to 0.0012°$
A1	half-ball separations shift in the +2.6% slip-throw direction; straight-in rows unchanged	separations -0.10 to $-0.13°$; straight-in rows bit-identical

Continued on next page

Table S3 *(continued)*

Expt.	Registered prediction	Measured outcome
A5, A6	gearing zero-crossings bit-identical (gearing zeroes slip); slip-branch ratios rescale by $(1+e_b)/(2e_b)$	zero-crossings bit-identical; transfer ratio $0.95007 \to 0.97508 = (1+e_b)/2$; half-ball sanity throw $4.5222 \to 4.6407°$
B1–B4	strike-only experiments unchanged	B1, B2, B4 bit-identical; B3 identical except a stale script stamp flag (bookkeeping)
D1–D3, F1–F3	D2, D3, F1 unchanged; F3 shifts only in its over-cut family (airborne contacts are slip-branch impacts); D1 shifts (post-impact slide is slip-branch)	confirmed as stated: D1 asymptote E_θ at 1 m/s $0.1307 \to 0.1729°$; F3 min-elevation curves and jump sweep bit-identical
E1, E3	compliant-cluster rows unchanged (force-based Coulomb consumes no impulse cap); the model contrast survives	E1 compliant slope bit-identical (contrast 0.139 vs 0.0295); E3 compliant squat rows bit-identical

S6. Presentation methods for the validated-scenario video artifacts

The artifact bundle's validated-scenario video pack (Section S2) renders sixteen catalog entries — each a solved strike, or a short ladder of strikes, replayed through the canonical binary shipped at the time of its render — as annotated slow-motion videos. The simulator's output of record is the kernel event stream, not any sampled rendering of it (main text, Section 3), and no claim of the paper rests on these videos: each entry first re-simulates its claimed scenario on that shipped canonical binary and must reproduce its pinned record digest *before* any frame is captured, so the videos are presen-

tations of already-verified records. The per-entry kernel stamp is recorded in the entry's gate report; the pack's stamps postdate the paper's battery pin along the supersession lineage of the main text, and each entry's digests are pinned against its own stamp. This section documents the deterministic time-remap and shutter synthesis used by those presentations, for two reasons: the frame schedule is deterministic — a pure function of the entry's playback plan, which is authored in the render tool's scenario catalog, with the derived schedule's summary (frame counts, worst rate-step ratio, shutter parameters, replay spans) pinned in the per-entry gate report — and the design choices are non-obvious enough that a reproducer would otherwise have to reverse-engineer them from the delivered bytes.

Frame schedule and the log-domain speed ramp.. Videos are delivered at a constant 60 frames per second; one output frame advances the kernel clock by $\Delta = 1/60$ s times a dimensionless playback rate r (kernel seconds consumed per output second; $r = 1$ is real time). A playback plan is a short list of rate segments over kernel time, each pair of consecutive segments separated either by a camera cut (a hard rate step on the cut frame) or by a *ramp*. A hard rate step without a cut reads as a speed jerk, so cut-less boundaries between rates r_1 and r_2 are ramped in the logarithmic rate domain:

$$r(u) \;=\; r_1 \left(r_2/r_1\right)^{S(u)}, \qquad S(x) \;=\; x^3\,(6x^2 - 15x + 10), \qquad u \in [0,1], \tag{S47}$$

where S is the quintic "smootherstep" polynomial, with zero first and second derivatives at both endpoints; the remap is therefore C^2 in log-rate, so the on-screen speed carries no visible acceleration kink. S is, term for term, the minimum-jerk rest-to-rest trajectory of human reaching movements (Flash

and Hogan, 1985, eq. (2), p. 1689): shaping the ramp with it makes the log-rate trajectory the minimum-jerk path between $\log_2 r_1$ and $\log_2 r_2$. The ramp duration follows a per-octave law,

$$D \;=\; \mathrm{clamp}\big(0.30 \cdot |\log_2(r_2/r_1)|,\; 0.5,\; 1.5\big) \text{ output seconds}, \tag{S48}$$

quantized to $n = \max(1, \mathrm{round}(D/\Delta))$ whole output frames — a fixed time cost per doubling of speed, bounded below and above so that small steps are not abrupt and large ones do not drag. The per-frame kernel advance integrates the ramp at the frame midpoint, $\mathrm{d}t_j = \Delta \cdot r((j + \frac{1}{2})/n)$ for $j = 0, \ldots, n-1$, and the ramp is scheduled as a *lead-in*: it starts on the last steady frame from which its integrated kernel consumption still completes at or before the segment boundary, so that whenever the outgoing segment can host the ramp — the common case — the target rate is established at or before the boundary and impacts land fully inside their slow-motion rate. A segment too short to host its ramp instead starts the ramp at segment entry and overshoots the boundary; the published pack contains one such replay pass, and the exception is why the rate-step law below and the per-entry gate windows are measured on the *emitted* clock, never on the nominal plan. Plan validation rejects non-positive or non-finite rates, which the log-domain form requires (the analogous restriction in the Web Audio specification is discussed below).

Perceptual basis for ramping in the log domain.. The ramp is built in the logarithmic speed domain, where equal steps are equal speed *ratios*; the quintic easing modulates how fast that axis is traversed (its rate reaching zero at both endpoints), so the constant-ratio character lives in the easing

variable $S(u)$ rather than literally in each output frame. The ratio axis is the perceptually relevant one for visual speed. The classical measurement is Weber-law behavior: the just-noticeable difference in speed grows in proportion to the reference speed (Brown, 1960). De Bruyn and Orban (1988) measured the Weber fraction for velocity discrimination at approximately 7% at the optimal speeds (84%-correct staircase criterion), roughly constant over a 4–64 deg/s plateau and rising (U-shaped) at slower and faster speeds. In modeling terms, Stocker and Simoncelli (2006) adopt a logarithmic speed representation for their likelihood and internal-noise functions (a log-Normal likelihood, $\tilde{v} = \ln((v + v_0)/v_0)$, in their Methods), motivated by exactly this Weber–Fechner character of speed discrimination — a modeling choice for the noise function, not a direct neural-coding proof; the neural grounding of a logarithmic, scale-invariant speed representation is the middle-temporal-area population analysis of Nover et al. (2005). A perceptually equal speed change is therefore an equal step in $\log r$, and Eq. (S47) ramps along exactly that axis, its minimum-jerk easing distributing the steps in time. The same reasoning is codified for audio in the W3C Web Audio API, whose `exponentialRampToValueAtTime` automation has the identical functional form $v(t) = V_0(V_1/V_0)^{(t-T_0)/(T_1-T_0)}$ and whose specification recommends exponential ramps for "filter frequencies and playback rate . . . because of the way humans perceive sound" (W3C Web Audio Working Group et al., 2021). The specification's domain restriction is instructive: an exponential ramp can neither reach nor pass through zero — the W3C text throws a `RangeError` only when the target value equals zero, while a zero or opposite-signed *start* value silently freezes the ramp (a ramp between two negative

values is permitted). Our plan validator is intentionally stricter than that domain: playback rates must be finite and strictly positive, so the degenerate cases the specification handles by freezing or throwing cannot be expressed in a plan at all.

Shutter synthesis over world-time windows.. Motion blur is synthesized, not post-processed: each delivered frame is the accumulation of 8 sub-frame renders spanning a fixed 310° shutter angle, i.e. the shutter opens at the frame's kernel time t_j and stays open for $(310/360)\cdot \mathrm{d}t_j$ of *kernel* (world) time, where $\mathrm{d}t_j$ is the integrated per-frame advance above — never the nominal segment rate. Blur length therefore follows the on-screen speed through ramps exactly as a physical camera filming the slow-motion playback would behave. In the frame domain the sub-frame animation steps sum to exactly Δ per output frame (the first sub-frame carries the closed-shutter remainder), so frame counts and presentation-layer animation lengths are independent of the shutter setting.

Monotone-map anchoring.. The schedule is emitted in monotone runs: within a run the map from output frame index to kernel time is strictly increasing, and a broadcast-style replay (a rewind to re-cover an impact) starts a new run, its boundary marked by a camera cut and a 0.4 s cross-dissolve identifying re-covered time. The *placement* of timed presentation content — predicted-overlay geometry, measured-value callouts, story-beat text reveals — is declared as *kernel-time* windows and resolved to output frames through this monotone map, never positioned by hand in output seconds; within a window, the reveal choreography itself (alpha fades, type entrances) plays authored fixed output-time durations — 0.4 s fades

matching the dissolve above — and a few entries carry small authored output-time leads. An annotation window that would match more than one run of its camera is rejected rather than guessed at. One mechanical law bounds the emitted clock: the worst consecutive cut-less per-frame kernel-advance ratio, measured on the emitted clock per monotone run, must not exceed 1.35×; the measured value is recorded in the per-entry gate report, and an entry that exceeds the bound fails before any frame is captured.

Relation to prior art.. The two ingredients of Eq. (S47) appear separately in published work. Speed-template curves in video-editing tools enumerate logarithmic and exponential playback-speed transitions among other shapes, motivated by editorial smoothness rather than psychophysics (Matias et al., 2017; Kuspa et al., 2012); and Weber–Fechner-guided playback-rate control appears in adaptive fast-forward summarization, where perceived fast-forward speed is modeled on a logarithmic axis (Jin et al., 2017). We have not found a published description that combines a perceptually motivated log-domain speed curve with C^2 minimum-jerk easing for cinematic slow-motion time remapping; we record the synthesis here as presentation methodology — it is not a claim of this paper, and none of the paper's results depend on it.

References

Alciatore, D.G., 2008. Pool and billiards physics principles by Coriolis and others. Self-hosted manuscript, https://drdavepoolinfo.com/physics/. Accessed 2026-07-21.

Alciatore, D.G., n.d.a. Frequently asked questions (FAQ) pages: Bank/kick systems, jump shots, throw effects. https://drdavepoolinfo.com/faq/. Accessed 2026-07-21.

Alciatore, D.G., n.d.b. High-speed video (HSV) catalog. https://drdavepoolinfo.com/high-speed-videos/. Individual clips cited by HSV number; accessed 2026-07-21.

Alciatore, D.G., n.d.c. Technical proofs (TP) catalog. https://drdavepoolinfo.com/technical_proofs/. Individual proofs cited by TP number; accessed 2026-07-21. Per-proof revision dates to be pinned at camera-ready.

Alciatore, D.G., n.d.d. TP A-14: Ball turn (throw) due to ball friction. https://drdavepoolinfo.com/technical_proofs/TP_A-14.pdf. Cut-induced-throw master curves; accessed 2026-07-21. Descriptive subtitle paraphrased; verbatim title to be pinned at camera-ready.

Alciatore, D.G., n.d.e. TP A-31: Cue ball deflection (squirt) and cue stick endmass. https://drdavepoolinfo.com/technical_proofs/TP_A-31.pdf. Endmass squirt closed form; accessed 2026-07-21. Descriptive subtitle paraphrased; verbatim title to be pinned at camera-ready.

Bouleau, N., 2006. Gaspard Coriolis et le jeu de billard comme auxiliaire scientifique. http://cermics.enpc.fr/~bouleaun/papiers/c51.pdf. Essay quoting Coriolis' final-direction robustness passage verbatim; accessed 2026-07-30.

Brown, R.H., 1960. Weber ratio for visual discrimination of velocity. Science 131, 1809–1810. doi:10.1126/science.131.3416.1809.

Cohen, G.L., 2002. Three cushion billiards: Notes on the diamond system. Sports Engineering 5, 43–51. doi:10.1046/j.1460-2687.2002.00095.x.

Coriolis, G.G., 1835. Théorie mathématique des effets du jeu de billard. Carilian-Goeury, Paris.

Cross, R., 2008. Cue and ball deflection (or "squirt") in billiards. American Journal of Physics 76, 205–212.

De Bruyn, B., Orban, G.A., 1988. Human velocity and direction discrimination measured with random dot patterns. Vision Research 28, 1323–1335. doi:10.1016/0042-6989(88)90064-8.

De Michele, C., 2022. Remark on algorithm 1010: Boosting efficiency in solving quartic equations with no compromise in accuracy. ACM Transactions on Mathematical Software 48, 46:1–46:3. doi:10.1145/3564270.

De Michele, C., 2024. Remark on algorithm 1010: Boosting efficiency in solving quartic equations with no compromise in accuracy. ACM Transactions on Mathematical Software 50, 1–3. doi:10.1145/3674833.

Flash, T., Hogan, N., 1985. The coordination of arm movements: an experimentally confirmed mathematical model. Journal of Neuroscience 5, 1688–1703. doi:10.1523/JNEUROSCI.05-07-01688.1985.

Gismeros Moreno, R., Corral, E., Meneses, J., Gómez García, M.J., Castejón, C., 2022. Modelling multiple-simultaneous impact problems with a nonlin-

ear smooth approach: Pool/billiard application. Nonlinear Dynamics 107, 1859–1886. doi:10.1007/s11071-021-07117-4.

Han, I., 2005. Dynamics in carom and three cushion billiards. Journal of Mechanical Science and Technology 19, 976–984.

Jin, H., Song, Y., Yatani, K., 2017. Elasticplay: Interactive video summarization with dynamic time budgets, in: Proceedings of the 25th ACM International Conference on Multimedia (MM '17), ACM. pp. 1164–1172. doi:10.1145/3123266.3123393.

Kiefl, E., 2024. Pooltool: A Python package for realistic billiards simulation. Journal of Open Source Software 9, 7301. doi:10.21105/joss.07301.

Kuspa, D., Davey, M., Warner, S., Young, P.E., 2012. Changing video playback rate. U.S. Patent 8,170,396 B2, Adobe Systems Inc. Application US 2008/0253735 A1, filed 2007. https://patents.google.com/patent/US20080253735A1/en. Accessed 2026-07-26.

Lankarani, H.M., Nikravesh, P.E., 1990. A contact force model with hysteresis damping for impact analysis of multibody systems. Journal of Mechanical Design 112, 369–376.

Lankarani, H.M., Nikravesh, P.E., 1994. Continuous contact force models for impact analysis in multibody systems. Nonlinear Dynamics 5, 193–207.

Leckie, W., Greenspan, M., 2006. An event-based pool physics simulator, in: Advances in Computer Games (ACG 2005), Springer. pp. 247–262.

Mathavan, S., Jackson, M.R., Parkin, R.M., 2010. A theoretical analysis of billiard ball dynamics under cushion impacts. Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science 224, 1863–1873. doi:10.1243/09544062JMES1964.

Matias, J.M., Sievert, O.K., Bodinet, J.E., 2017. Variable playback speed template for video editing application. U.S. Patent 9,679,605 B2, GoPro, Inc. Filed 2015; issued 13 June 2017. https://patents.google.com/patent/US9679605B2/en. Accessed 2026-07-26.

Nover, H., Anderson, C.H., DeAngelis, G.C., 2005. A logarithmic, scale-invariant representation of speed in macaque middle temporal area accounts for speed discrimination performance. Journal of Neuroscience 25, 10049–10060. doi:10.1523/JNEUROSCI.1661-05.2005.

Orellana, A.G., De Michele, C., 2020. Algorithm 1010: Boosting efficiency in solving quartic equations with no compromise in accuracy. ACM Transactions on Mathematical Software 46, 20:1–20:28. doi:10.1145/3386241.

Shepard, R., 2001. Everything you always wanted to know about cue ball squirt, but were afraid to ask. Self-published manuscript; archived at https://drdavepoolinfo.com/physics/. Accessed 2026-07-21.

Stocker, A.A., Simoncelli, E.P., 2006. Noise characteristics and prior expectations in human visual speed perception. Nature Neuroscience 9, 578–585. doi:10.1038/nn1669.

Stronge, W.J., 2018. Impact Mechanics. 2nd ed., Cambridge University Press.

W3C Web Audio Working Group, Adenot, P., Choi, H., 2021. Web audio API. World Wide Web Consortium (W3C) Recommendation, 17 June 2021, https://www.w3.org/TR/webaudio-1.0/. This version https://www.w3.org/TR/2021/REC-webaudio-20210617/. The exponentialRampToValueAtTime() formula, the RangeError-on-zero-value rule, and the exponential-ramp perceptual rationale are unchanged in the successor Web Audio API 1.1 First Public Working Draft (5 Nov 2024, https://www.w3.org/TR/webaudio-1.1/). Accessed 2026-07-26.